

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Answer:

- When compared to other seasons, the demand of bicycles is lower in the month of spring.
- In comparison to 2018, the demand for bicycles grew in 2019.
- The months of June to September are the busiest for bike sales. The month of January has the lowest demand.
- Bike demand is lower over the holidays than when it is not.
- During the weekdays, the demand for bicycles is about identical.
- There is no discernible difference in bike demand between working and nonworking days.
- When the weather is clear and there are few clouds, bike demand is strong; nevertheless, demand is lower when there is light snow or light rain. We don't have any information about Heavy Rain + Ice Pellets + Thunderstorm + Mist, Snow + Fog at this time.

2. Why is it important to use `drop_first=True` during dummy variable creation?

Answer:

Let's look at an example where you would obtain a redundant feature if you didn't use "drop first." If you have the feature "Is male," and you use "get dummies," you'll get two features: "Is male 0" and "Is male 1," but they're both redundant; you only need one of them, and the other will be the exact opposite of the other. For Example

row	is_male
0	1
1	0

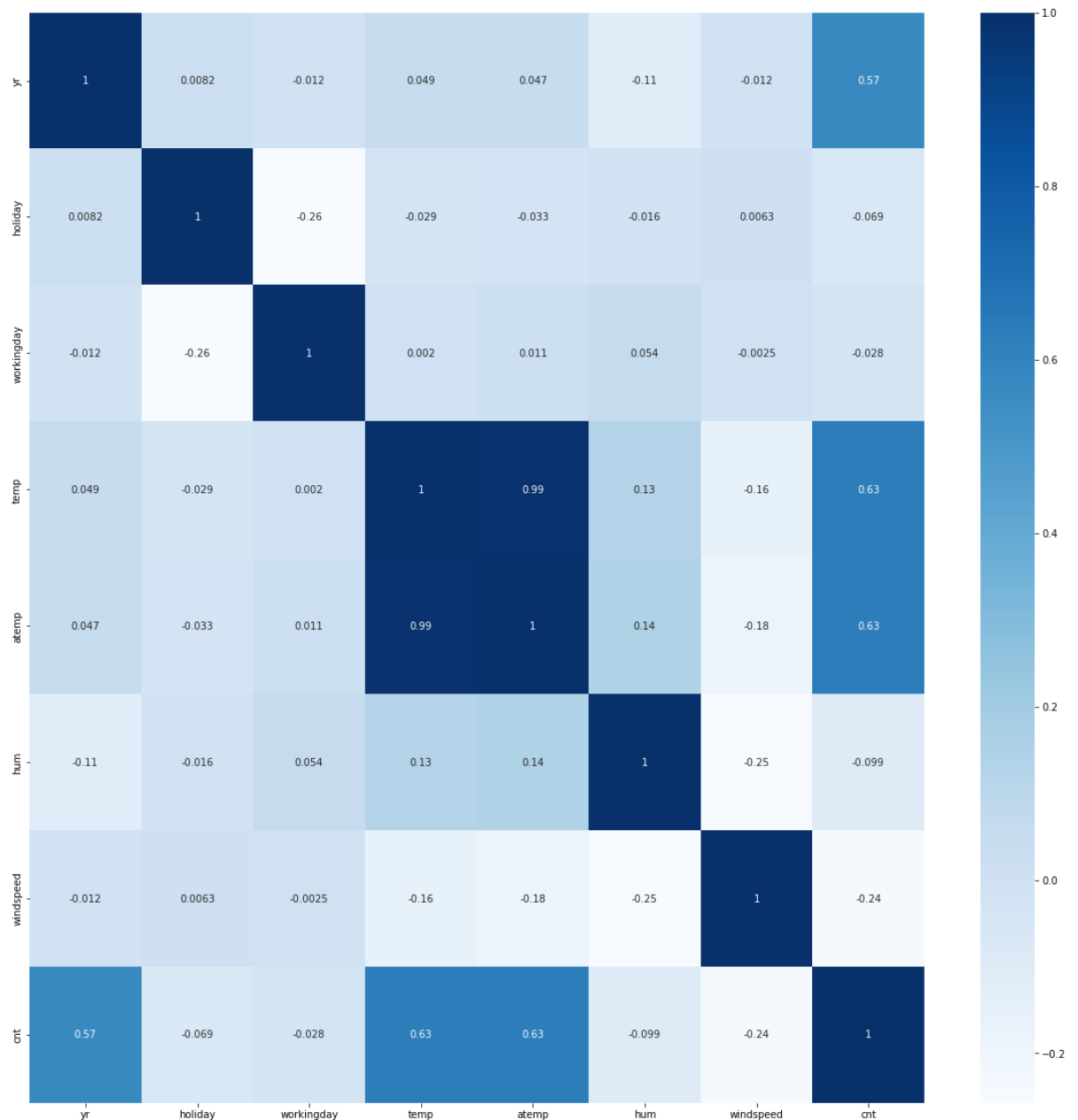
after applying `get_dummies`

row	is_male_0	is_male_1
0	0	1
1	1	0

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable? (1 mark)

Answer

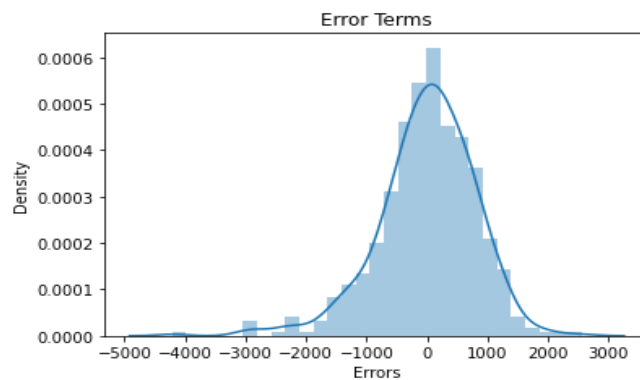
Temperature



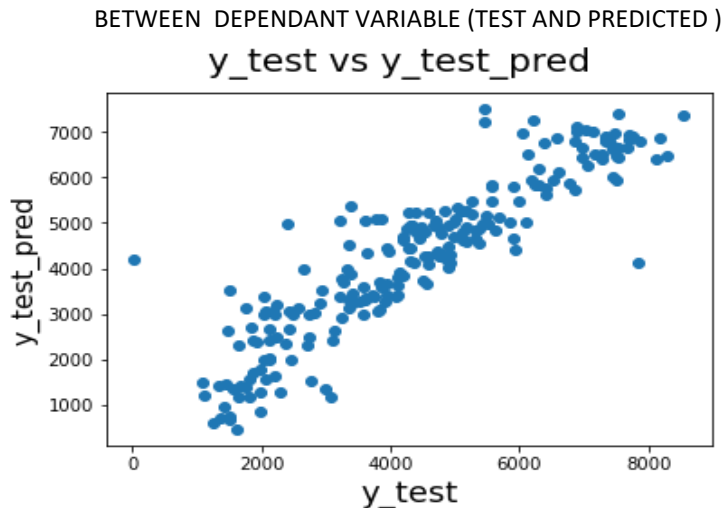
4. How did you validate the assumptions of Linear Regression after building the model on the training set? (3 marks)

Answer:

- Residual errors follow normal distribution



- Maintains linear relation



5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes? (2 marks)

Answer:

As per our final Model, the top 3 predictor variables that influences the bike booking are:

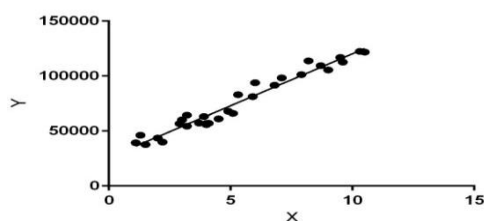
- Temperature (temp) - A coefficient value of '3576.4598' indicated that a unit increase in temp variable increases the bike hire numbers by 0.5636 units.
- Season_spring - A coefficient value of '-1217.6811' indicated that, w.r.t Weathersit1, a unit increase in Weathersit3 variable decreases the bike hire numbers by 1217.6811 units.
- Bad_weather - A coefficient value of '-2317.0972' indicated that a unit increase in yr variable increases the bike hire numbers by -2317.0972 units.

General Subjective Questions

1. Explain the linear regression algorithm in detail.

Answer:

Linear regression is a supervised learning machine learning algorithm. It carries out a regression task. Based on independent variables, regression models a goal prediction value. It is mostly utilised in forecasting and determining the link between variables. Different regression models differ in terms of the type of relationship they evaluate between dependent and independent variables, as well as the amount of independent variables they employ.



Linear regression is used to predict the value of a dependent variable (y) based on the value of an independent variable (x). As a result of this regression technique, a linear relationship between x (input) and y (output) is discovered (output). As a result, the term Linear Regression was coined. In the diagram above, X (input) represents job experience and Y (output) represents a person's wage. For our model, the regression line is the best fit line.

We are given the following model to train:

(univariate - one input variable(parameter)) x: input training data

y: the data's labels (supervised learning)

It fits the best line to predict the value of y for a given value of x when training the model. By determining the best 1 and 2 values, the model obtains the best regression fit line.

1st: interception

2: the x-coefficient

We get the best fit line after we get the best 1 and 2 values. So, when we use our model to predict the value of y for the input value of x, it will predict the value of y.

How can the 1 and 2 values be updated to acquire the greatest fit line?

(J) Cost Function:

The best-fit regression line is obtained by obtaining the best-fit regression line.

$$\text{minimize } \frac{1}{n} \sum_{i=1}^n (\text{pred}_i - y_i)^2$$

The Root Mean Squared Error (RMSE) between the predicted y value (pred) and the true y value (true) is the cost function(J) of Linear Regression (y).

Descent via Gradient:

Gradient Descent is used to update 1 and 2 values in order to minimise Cost function (minimising RMSE value) and get the best fit line. The goal is to start with random 1 and 2 numbers and iteratively update the values until the cost is as low as possible.

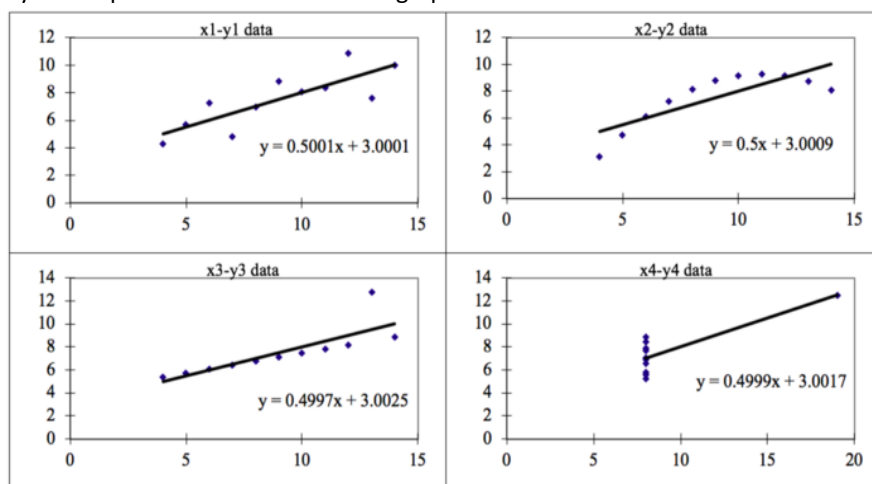
2. Explain the Anscombe's quartet in detail.

Answer:

Anscombe's quartet consists of four data sets with virtually equal simple descriptive statistics but radically varied distributions and graphing appearances.

This emphasises the importance of visualising data before applying various algorithms to create models from it, implying that data features must be plotted in order to see the distribution of samples, which can aid in identifying various anomalies in the data such as outliers, data diversity, linear separability, and so on. Furthermore, Linear Regression can only be used to fit data with linear connections and is incapable of handling any other datasets.

When statistical information for all datasets is roughly comparable and models are shown on a scatter plot, each dataset produces a unique plot that is uninterpretable by any regression algorithm deceived by these quirks. Consider these four graphs from a random dataset.



The four datasets are as follows:

1. Dataset 1: this is a good fit for the linear regression model.

2. Dataset 2: due to the non-linear nature of the data, a linear regression model could not be fit adequately.
3. Dataset 3: displays the dataset's outliers that the linear regression model cannot manage.
4. Dataset 4: displays the dataset's outliers that the linear regression model cannot manage.

3. What is Pearson's R?

Pearson's r is a numerical representation of the strength of a linear relationship between two variables. The correlation coefficient will be positive if the variables tend to go up and down together. The correlation coefficient will be negative if the variables tend to go up and down in opposite directions, with low values of one variable correlated with high values of the other.

The Pearson's correlation coefficient ranges from -1 to +1, depending on the following factors:

- The data is fully linear with a positive slope if $r = 1$. (i.e., both variables tend to change in the same direction)
- The data is fully linear with a negative slope if $r = -1$. (i.e., both variables tend to change in different directions)
- There is no linear relationship if $r = 0$.
- $r > 0.5$ indicates a poor relationship.
- $r > 0.8$ denotes a moderate relationship.
- $r > 0.9$ denotes a strong correlation.

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling is a data pre-processing technique that is used to independent variables in order to normalise the data within a given range. It also aids in the acceleration of algorithmic calculations.

Scaling is performed as majority of the time, the acquired data set contains features with a wide range of magnitudes, units, and ranges. If scaling is not done, the algorithm will only consider magnitude rather than units, resulting in erroneous modelling. To solve this problem, we must scale all of the variables to the same magnitude level.

Scaling only changes the coefficients and not the other parameters such as the t-statistic, F-statistic, p-values, R-squared, and so on.

$$\text{Min max scaling}(x) = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardization Scaling:

- Standardization replaces the values by their Z scores. It brings all of the data into a standard normal distribution which has mean (μ) zero and standard deviation one (σ).
Standardisation(x) = $\frac{x - \text{mean}}{\text{standard deviation}(x)}$
- `sklearn.preprocessing.scale` helps to implement standardization in python.
- One disadvantage of normalization over standardization is that it loses some information in the data, especially about outliers.

S.NO.	Normalization	Standardization
1.	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2.	It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
3.	Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
4.	It is really affected by outliers.	It is much less affected by outliers.
5.	Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.
6.	This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.
7.	It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
8.	It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Answer

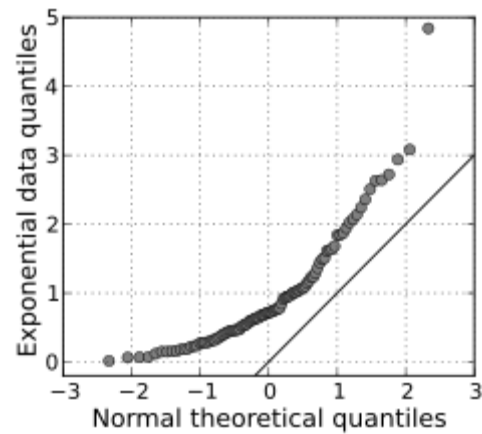
VIF = infinity if there is perfect correlation. This demonstrates that two independent variables have a perfect correlation. We get $R^2 = 1$ in the event of perfect correlation, which leads to $1/(1-R^2)$ infinite. To overcome this issue, we must remove one of the variables that is producing the perfect multicollinearity from the dataset.

An infinite VIF value suggests that a linear combination of other variables can exactly express the related variable (which show an infinite VIF as well).

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Answer:

Q-Q charts (Quantile-Quantile plots) are plots that compare two quantiles. A quantile is a percentage of the population in which specific values fall below it. The median, for example, is a quantile where 50% of the data falls below it and 50% of the data falls above it. Q Q plots are used to determine whether two sets of data are from the same distribution. On the Q Q plot, a 45 degree angle is drawn; if the two data sets are from the same distribution, the points will fall on that reference line.



The 45 degree reference line on a Q Q plot:

The points in the Q–Q plot will roughly lie on the line $y = x$ if the two distributions being compared are similar. The points in the Q–Q plot will roughly lie on a line if the distributions are linearly connected, but not necessarily on the line $y = x$. Q–Q plots can also be used to estimate parameters in a location-scale family of distributions graphically.

A Q–Q plot is used to compare the morphologies of two distributions, offering a graphical representation of how features like location, scale, and skewness differ in the two distributions.